

Researchers: Oldest Evidence of Lead Pollution Found in Greece

Ancient Greece produced the earliest records of democracy - and, scientists now say, lead pollution.

That is the finding of a study that recently appeared in *Communications Earth and Environment*. The researchers studied sediment cores removed from mainland Greece and below the Aegean Sea. They found in the cores the oldest known evidence of lead pollution in the environment dating back around 5,200 years.

That is 1,200 years older than the former oldest evidence of lead pollution. That evidence was discovered in Serbia.

In ancient times, lead was released into the atmosphere as a product of copper and silver production. The toxic, or poisonous, metal turned to dust over time and settled onto the soil.

Joseph Maran of Heidelberg University was a co-writer of the new study.

“Silver was used for jewelry, for special objects — but it wasn't found in a pure state,” it was mined in ore combined with lead, Maran said

The place with the earliest signs of lead is in northeastern Greece, near the island of Thasos. Past evidence suggests Thasos was one of the area’s most important places for silver mining and metalwork, said Maran.

Yale historian Joseph Manning, who was not involved in the study, said “Lead released from smelting is the world’s first form of toxic or industrial pollution.”

This undated handout photo shows the pickup of sediment cores off Peloponnese in the Aegean Sea onboard the Meteor vessel. (Andreas Koutsodendris via AP)

The researchers found that levels of lead contamination remained fairly low in ancient Greece. The levels remained low throughout the Bronze Age, the Classical period and the Hellenistic period. The Classical period is famous for Athenian democracy, and the thinkers Socrates and Plato. And the Hellenistic period saw Greek cultural influence reach its height across the Mediterranean.

But the new study shows that around 2,150 years ago “a very strong and abrupt increase” in lead releases caused by human activities took place across Greece, said co-writer Andreas Koutsodendris of Heidelberg University.

Around that time, the Roman army won control of the Greek peninsula, completely changing the area's society and economy. As Roman trade, colonies and shipping grew across the Mediterranean Sea and Black Sea, demand for silver coins also grew. As a result, more smelting was required, Koutsodendris said.

Later the Roman Empire used lead to make tools for eating and building, including pipes.

Past research – including a study of ice cores from Greenland – had detected high levels of lead across much of the Northern Hemisphere during Roman times.

But the new study adds a more exact “and local picture to how lead levels changed,” said Nathan Chellman. Chellman, an environmental scientist at the University of Nevada, Reno, was not involved in the research.

I'm John Russell.

Christina Larson reported on this story for the Associated Press. John Russell adapted it for VOA Learning English.

Vocabulary

sediment core – n. ตัวอย่างชั้นดินทรงกระบอกที่ให้ข้อมูลเกี่ยวกับสภาพพื้นผิวเมื่อเวลาผ่านไป

smelt – v. หลอมหรือหลอมรวมสารเพื่อแยกโลหะ

abrupt – adj. ทันทีทันใดและไม่คาดคิด

Reading Comprehension Quiz

1. What two things are mentioned as being produced by Ancient Greece, according to scientists?

- A) Democracy and philosophy
- B) Records of democracy and lead pollution
- C) Copper and silver production
- D) Jewelry and special objects

2. What was the primary method used by researchers to find evidence of lead pollution?

- A) Analyzing ancient artifacts
- B) Studying historical documents
- C) Examining sediment cores
- D) Excavating ancient mines

3. How far back does the *oldest known evidence* of lead pollution found in this study date?

- A) Around 1,200 years ago
- B) Around 2,150 years ago
- C) Around 5,200 years ago
- D) Around 3,000 years ago

4. Where was the *previous* oldest evidence of lead pollution discovered, before this new study?

- A) Mainland Greece
- B) The Aegean Sea
- C) Serbia
- D) Thasos

5. According to the article, how was lead initially released into the environment in ancient times?

- A) Through the use of lead pipes for water.
- B) As a byproduct of copper and silver production.
- C) From volcanic eruptions.
- D) By burning lead for fuel.

6. Why was lead often associated with silver production in ancient times?

- A) Lead was used to polish silver jewelry.
- B) Silver was typically found in ore combined with lead.
- C) Lead was a cheaper alternative to silver.
- D) Miners used lead tools to extract silver.

7. Where in Greece did researchers find the earliest signs of lead pollution?

- A) Near the Peloponnese
- B) In the Aegean Sea
- C) Northeastern Greece, near Thasos
- D) Athens

8. What did Joseph Manning, a Yale historian, state about lead released from smelting?

- A) It was the most dangerous form of ancient pollution.
- B) It marked the beginning of industrial pollution.
- C) It primarily affected only the local environment.

D) It was unrelated to economic growth.

9. During which historical periods did lead contamination levels remain relatively low in ancient Greece?

- A) Only during the Bronze Age
- B) Throughout the Bronze Age, Classical period, and Hellenistic period
- C) After the Roman army gained control
- D) Only during the time of Socrates and Plato

10. What significant cultural development characterized the Hellenistic period, as mentioned in the text?

- A) The invention of democracy.
- B) The rise of philosophers like Socrates.
- C) Greek cultural influence reaching its peak across the Mediterranean.
- D) The beginning of widespread lead usage.

11. According to Andreas Koutsodendris, what significant change in lead releases occurred approximately 2,150 years ago?

- A) Lead pollution completely ceased.
- B) There was a very strong and abrupt increase in lead releases.
- C) Lead levels began to decrease slowly.
- D) Scientists began to monitor lead levels.

12. What historical event is linked to the "strong and abrupt increase" in lead releases around 2,150 years ago?

- A) The invention of new mining techniques.
- B) The discovery of lead in new regions.
- C) The Roman army winning control of the Greek peninsula.
- D) The start of the Classical period.

13. Why did the demand for silver coins increase during Roman expansion?

- A) Silver became a new form of currency.
- B) Roman trade, colonies, and shipping grew.
- C) The Roman army required more pay.
- D) Greece adopted the Roman currency system.

14. Besides silver production, for what other purposes did the Roman Empire use lead?

- A) As a fertilizer for agriculture.

- B) To create artwork and sculptures.
- C) To make tools for eating and building, including pipes.
- D) For medicinal purposes.

15. What kind of past research supported the idea of high lead levels across the Northern Hemisphere during Roman times?

- A) Geological surveys in the Mediterranean.
- B) Studies of ancient Roman city ruins.
- C) Analysis of ice cores from Greenland.
- D) Historical texts from Roman historians.